VIKNESH.G 23MID0161

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In [3]:
```
import pandas as pd
import numpy as np
```

CREATE THE DATAFRAME

In [5]:
```
import pandas as pd

data = {
   "RANK": [
        1, 2, 3, 4, 5, 6, 7, 8, 9, 10,
        11, 12, 13, 14, 15, 16, 17, 18, 19, 20,
        21, 22, 23, 24, 25, 26, 27, 28, 29, 30
    ],
    "CITY": [
        "Tampa", "Breda", "Singapore", "Oslo", "Sydney", "Vancouver", "Montreal", "Cambridge", "Xalapa", "Durban",
        "Johannesburg", "Sacramento", "Frankfurt", "Geneva", "Veracruz", "Amsterdam", "Seattle", "Toronto", "Miami"
        "Tel Aviv", "Guadalajara", "León", "Turin", "Los Angeles", "Buenos Aires", "Franca", "New York", "Cape Town
    ],
    "COUNTRY": [
        "USA", "NLD", "SGP", "NOR", "AUS", "CAN", "CAN", "USA", "MEX", "ZAF",
        "ZAF", "USA", "DEU", "CHE", "MEX", "NLD", "USA", "CAN", "USA", "USA",
        "ISR", "MEX", "MEX", "ITA", "USA", "ARG", "BRA", "USA", "ZAF", "GBR"
    ],
    "GVI": [
        36.1, 29.3, 29.3, 28.8, 25.9, 25.9, 25.5, 25.3, 24.0, 23.7,
        23.6, 23.6, 21.5, 21.4, 21.4, 20.6, 20.0, 19.5, 19.4, 18.2,
        17.5, 17.3, 16.5, 16.2, 15.2, 14.5, 13.7, 13.5, 13.4, 12.7
    ]
}
```

```
df = pd.DataFrame(data)
print(df)
```

```
    RANK          CITY COUNTRY   GVI
0      1         Tampa     USA  36.1
1      2         Breda     NLD  29.3
2      3     Singapore     SGP  29.3
3      4          Oslo     NOR  28.8
4      5        Sydney     AUS  25.9
5      6     Vancouver     CAN  25.9
6      7      Montreal     CAN  25.5
7      8     Cambridge     USA  25.3
8      9        Xalapa     MEX  24.0
9     10        Durban     ZAF  23.7
10    11  Johannesburg     ZAF  23.6
11    12    Sacramento     USA  23.6
12    13     Frankfurt     DEU  21.5
13    14        Geneva     CHE  21.4
14    15      Veracruz     MEX  21.4
15    16     Amsterdam     NLD  20.6
16    17       Seattle     USA  20.0
17    18       Toronto     CAN  19.5
18    19         Miami     USA  19.4
19    20        Boston     USA  18.2
20    21      Tel Aviv     ISR  17.5
21    22   Guadalajara     MEX  17.3
22    23          León     MEX  16.5
23    24         Turin     ITA  16.2
24    25   Los Angeles     USA  15.2
25    26  Buenos Aires     ARG  14.5
26    27        Franca     BRA  13.7
27    28      New York     USA  13.5
28    29     Cape Town     ZAF  13.4
29    30        London     GBR  12.7
```

1. avg gvi of the usa

method 1 (prompt: how to identify a aggregated function of the p[articular repeatation of value in a col)

In [8]:
```
target='USA'
avg_usa= df[df['COUNTRY']==target]['GVI'].mean()
print("avg gvi of usa : {}".format(avg_usa))
```

```
avg gvi of usa : 21.4125
```

method 2 using pivot_table (prompt : give me thesyntax of the pivot table and what it does)

In [9]:
```
pivot = df.pivot_table(
    index="COUNTRY",
    values="GVI",
    aggfunc=["mean", "count"]
)
print(pivot)
```

```
              mean count
               GVI   GVI
COUNTRY
ARG      14.500000     1
AUS      25.900000     1
BRA      13.700000     1
CAN      23.633333     3
CHE      21.400000     1
DEU      21.500000     1
GBR      12.700000     1
ISR      17.500000     1
ITA      16.200000     1
MEX      19.800000     4
NLD      24.950000     2
NOR      28.800000     1
SGP      29.300000     1
USA      21.412500     8
ZAF      20.233333     3
```

In [23]:
```
target = "USA"
df2 = pivot.loc[[target]]
print(df2)
```

```
           mean count
            GVI   GVI
COUNTRY
USA      21.4125     8
```

method 3

In [28]:
```
usa_mean = pivot.loc["USA", ("mean", "GVI")]
print(f"USA Mean: {usa_mean:.4f}")
```

```
USA Mean: 21.4125
```

2. minimum gvi of netherlands

method 1

In [30]:
```
target='NLD'
min_nld= df[df['COUNTRY']==target]['GVI'].min()
print("min value of nld : {}".format(min_nld))
```

```
min value of nld : 20.6
```

method 2 (prompt : can i find the min of the row with target value using loc)

In [32]:
```
target = "NLD"
min_nld = df.loc[df["COUNTRY"] == target, "GVI"].min()
print(f"min value of nld : {min_nld}")
```

```
min value of nld : 20.6
```

method 3

In [33]:
```
country_minimums = df.groupby("COUNTRY")["GVI"].min()
min_nld = country_minimums[target]
print(f"min value of nld : {min_nld}")
```

```
min value of nld : 20.6
```

3. sum of gvi of all cities in italy

method 1

In [38]:
```
italy_total_gvi = df.loc[df["COUNTRY"] == "ITA", "GVI"].sum()
print(f"Total GVI for Italy: {italy_total_gvi}")
```

```
Total GVI for Italy: 16.2
```

method 2 using loc

In [37]:
```
italy_total_gvi = df[df["COUNTRY"] == "ITA"]["GVI"].sum()
print(f"The total sum of GVI for all cities in Italy is: {italy_total_gvi}")
```

```
The total sum of GVI for all cities in Italy is: 16.2
```

method 3 (using gvi)

In [39]:
```
country_sums = df.groupby("COUNTRY")["GVI"].sum()
italy_total_gvi = country_sums["ITA"]
print(f"Total GVI for Italy: {italy_total_gvi}")
```

```
Total GVI for Italy: 16.2
```

4. find the city and country with least gvi

method 1

In [41]:
```
min_index = df["GVI"].idxmin()
lowest_city = df.loc[min_index, "CITY"]
lowest_country = df.loc[min_index, "COUNTRY"]
lowest_gvi = df.loc[min_index, "GVI"]
print(f"Least GVI: {lowest_city}, {lowest_country} ({lowest_gvi})")
```

```
Least GVI: London, GBR (12.7)
```

method 2 (prompt : what else method to find ,can i find using iloc)

In [42]:
```
df_sorted = df.sort_values(by="GVI", ascending=True)
lowest_city = df_sorted.iloc[0]["CITY"]
lowest_country = df_sorted.iloc[0]["COUNTRY"]
lowest_gvi = df_sorted.iloc[0]["GVI"]
print(f"Least GVI: {lowest_city}, {lowest_country} ({lowest_gvi})")
```

```
Least GVI: London, GBR (12.7)
```

method 3 (nsmallest)

In [ ]:
```
least_gvi_row = df.nsmallest(1, "GVI")
lowest_city = least_gvi_row["CITY"].values[0]
lowest_country = least_gvi_row["COUNTRY"].values[0]
lowest_gvi = least_gvi_row["GVI"].values[0]
print(f"Least GVI: {lowest_city}, {lowest_country} ({lowest_gvi})")
```